Sahil Vora

Software Developer
Seattle, USA

Email: sahilvora2024@gmail.com
Phone:+1 (602) 919-9535 **LinkedIn:** sahil-vora

Education

- **Arizona State University** **May 2024**
  Master of Science - Computer Science w/ Thesis on Computer Vision for Medical Imaging GPA: 4.00/4.00
- **Manipal Institute of Technology, Manipal, India** **July 2020**
  B.Tech in Computer and Communication Engineering w/ specialization in Big Data GPA: 8.69/10.0

Technical Skills

- **Programming**: Python (proficient), Java (advanced), C++ (intermediate), SQL (intermediate)
- **ML Frameworks**: PyTorch (proficient), SciKit-Learn (intermediate), TensorFlow (intermediate)
- **Platforms & DevOps**: Linux, AWS, Amazon Bedrock & Sagemaker, Git, Jenkins, Docker

Work Experience

- **Software Developer: Amazon Web Services**, Seattle, WA *(Dec'24-Present)*
  - Maintain and scale Tier-0 infrastructure for AWS Key Management Service, automating regional and zonal build workflows to enhance platform reliability and availability. (helping in 5 new regions and 4 AZs and reducing manual 100+ engineering hours)
  - Lead migration of KMS tools, infrastructure, and operational workflow ticketing systems to a more secure, scalable, and efficient platform.
  - Design and implement agentic AI solutions using Amazon Bedrock and Q to assist operators, reduce manual toil, and improve operational efficiency for operational ticketing and helping in cleanup deployment failures in 20+ pipelines.
- **Technology Developer: Barclays**, Whippany, NJ *(Jul'24-Dec'24)*
  - Building a deep generative model for website clickstream analytics using Amazon SageMaker and Bedrock to enhance real-time customer behavior predictions and support.
  - Implementing dynamic config management for a Kafka-based platform, improving CI/CD pipelines with GitLab and Chef.
- **Research Assistant: Arizona State University w/ Mayo Clinic**, Tempe, AZ *(Jan'23-May'24)*
  - Researched and developed unsupervised denoising and super-resolution of medical imaging using diffusion models under the supervision of Dr. Baoxin Li, achieving state-of-the-art results and contributing to 5 publications.
  - Conducted in-depth research on low-shot generalization, creating multi-modal agents for context and feature aggregation, introducing innovative methodologies to enhance data efficiency under challenging computer vision conditions.
- **Software Developer: Sabre**, Bangalore, India *(Oct'21-Jul'22)*
  - Led the migration of data platforms to Google Cloud with 100% success, optimizing APIs and automating pipelines for legacy system integration.
- **Associate Software Developer: Publicis Sapient**, Gurugram, India *(Jan'20-Oct'21)*
  - Maintained zero downtime for a gaming e-commerce platform during peak sales and increased revenue by 50% through feature deployment and performance optimization using Google Analytics and Salesforce Commerce Cloud.

Relevant Publications

- **S. Vora**, R. Paul, P.L.K. Ding, A. Patel, L. Hu, R. Sharpe, Y. Zhou, B. Li. MEDL: Unsupervised Multi-Stage Ensemble Deep Learning Using Diffusion Models for Denoising MRI Scans. *IEEE 22nd International Symposium on Biomedical Imaging (ISBI)*, 2025 *doi:10.1109/isbi60581.2025.10981026*
- R.Paul, **S. Vora**, P.L.K. Ding, A. Patel, L. Hu, B. Li, Y. Zhou. Transferable Variational Feedback Network for Vendor and Sequence Generalization in Accelerated MRI. *IEEE Big Data Mining and Analytics*, 2025. *doi:10.26599/bdma.2025.9020013*
- R.Paul, **S. Vora**, N. Thakur, B. Li. A-FSL: Adaptive Few-Shot Learning via Task-Driven Context Aggregation and Attentive Feature Refinement. *International Conference on Pattern Recognition*, 2024. *doi:10.1007/978-3-031-78395-1_7*
- R.Paul, **S. Vora**, K.P.L. Ding, A. Patel, L. Hu, B. Li, Y. Zhou. Transferable Variational Feedback Network for Vendor Generalization in Accelerated MRI. *International Conference on AI in Healthcare*, 2024. *doi:10.1007/978-3-031-67285-9_4*
- R.Paul, **S. Vora**, B. Li. Instance Adaptive Prototypical Contrastive Embedding for Generalized Zero-Shot Learning. *International Joint Conference on Artificial Intelligence (IJCAI)*, 2023. arXiv:2309.06987

Projects

- **Echo Chamber Detection in Social Media** (*Python, Pytorch, Gephi, HuggingFace*) - Developed echo chamber detection algorithms for social media using network & content features, measuring user timeline similarity through tweet embedding with SBERT & NLP, combating misinformation and revealing political polarization. [Link]
- **COVID-19 Vaccine Stance Detection** (*Python, MATLAB, Data Processing, Data Analytics*) - Conducted a Social Media exploratory project to analyze public attitudes toward COVID-19 vaccines, utilizing hashtag-based data from both Pro and Anti-vaccine campaigns to identify cluster formations and campaign epicentres. [Link]

Honors and Awards

- Best Toil Reduction Award, AWS Crypto Essentials Hackathon 2025
- 2025 Seattle Half Marathon Finisher 2025
- Best Paper Award, International Joint Conference on Artificial Intelligence (IJCAI): GLOW 2023
- University Winner, Flipkart Grid - Teach the Machines Machine, Learning Challenge 2019